Lecture 5

Tuesday, 5 September 2023 10:32 AM

Crystal lattice Structures (XRD, bulk characterization)

→ Next atom neighbors, → affects interactions w/ other atoms

→ Crystalline → regular order (zeolite)

Amorphous → No repeat unit; local disorder; glass, sand.

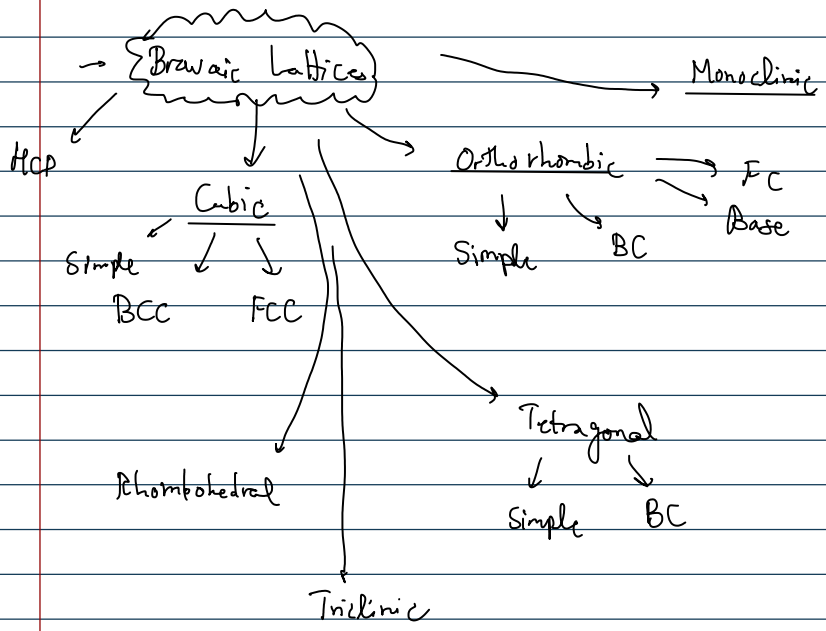
SiO_2 thin-film on Ru(0001) → first STM image of amorphous SiO_2

→ Study of active sites on amorphous silica is a challenge

→ lattice

→ 14 possible arrangements of points in space possible

→ Surface termination → a specific bulk arrangement will always end/ have specific surface termination



Axis are defined along the unit cell edges (and not cartesian)

[uvw] → no. of lattice steps taken to reach desired points

↳ These are vectors
so can be reduced to that

↓

But not that useful (we care about surfaces)

↓

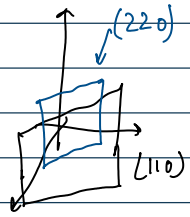
Crystal plane have <u>miller indices</u>

(hkl)

→ Intersection (has a value) of to axis

→ Parallel to an axis (has 0 value)

} Just is inverse Lie.



Hexagonal symmetry: Miller-Bravais (hkil)

$i = -(h+k)$

Maximize Coordination no. in bulk (FCC, HCP) ; Is Au, Pt FCC?

↓

?

→ Linear density

Planar density